ECE Study Assistant - capstone report

A LangGraph retrieval-augmented agent over eight lecture-note documents. Tasks 4.1-4.4.

Run configuration: **offline** generation, **sentence-transformers/all-MiniLM-L6-v2** embeddings, chunk 800/150, top-k 4. Generated 2026-10-12. Reproducible with no API key.

1. Overview

The system answers three kinds of question: those grounded in the notes (retrieve, grade, answer with a citation), those needing a calculation or formula (a tool call), and small talk. It runs with no API key using a deterministic extractive generator and upgrades to Groq or Gemini when a key is present. The graph is deliberately conditional and cyclic - a failed relevance grade rewrites the query and retries retrieval - which is the mechanism the evaluation measures.

2. Retrieval (4.1)

Pipeline: PyPDFLoader -> RecursiveCharacterTextSplitter(800/150) -> all-MiniLM-L6-v2 (384-d, L2-normalised) -> FAISS IndexFlatIP. The corpus is 8 generated documents, 27 chunks, mean 652 chars. Flat inner product over normalised vectors gives exact cosine similarity, so the score in every trace is a real cosine in [-1, 1].

3. Agent workflow (4.2)

The agent is a LangGraph state machine with a MemorySaver checkpointer. Routing is a runtime decision and a failed grade loops back through query rewriting; a linear chain cannot express that. The LLM layer exposes four decisions (route, grade, rewrite, generate) rather than a generic prompt call, which is what lets the whole system run and be evaluated with no key. Tools are offline: a calculator behind a restricted AST walk (never eval) and a curated, sourced formula table.

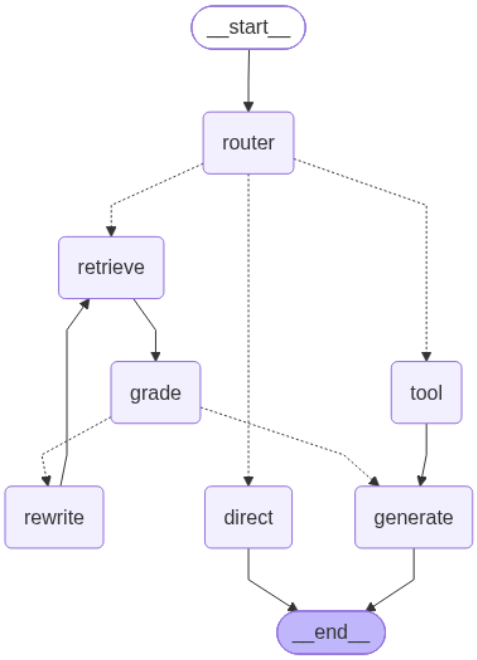


Figure 1. Topology, rendered from the compiled graph.

4. Evaluation (4.3)

22-question labelled set, run twice. **Offline** (no key): latency median 56.0 ms, p95 68.4 ms. **LLM** (Groq openai/gpt-oss-20b): median 4405.1 ms, model time median 3377.8 ms. Both runs are tagged and sent to the week4-study-assistant LangSmith project (22 trees each), and replayed through the week4-study-assistant-eval dataset + evaluators; docs/langsmith_figures.py renders the live trees to docs/traces/langsmith-*.png.

Metric	offline	LLM (Groq)
routing accuracy	100%	100%
retrieval hit rate	100%	100%
answer keyword rate	93%	100%
tool success rate	100%	100%
out-of-scope refusal rate	67%	100%

The three 100% metrics are a ceiling from a small, self-authored set - the two below 100% are the informative results. **q20 (Friis equation)**: absent from the corpus, but 'transmission ... formula' passes lexical relevance grading, so the agent gives a fluent, cited, wrong answer - the case for a generative grader (the Groq grader does catch it). **q22 (FM bandwidth)**: retrieval was correct; the extractive generator quoted the neighbouring sentence instead of Carson's rule - a generation failure, not a retrieval one.

4.1 Ablations

chunk / overlap	chunks	hit@4	top-1 cos
400 / 100	48	100%	0.593
800 / 150	27	100%	0.553
1200 / 240	17	100%	0.517

Retry policy: the low-coverage floor drops unanswerable questions from **8.0 to 4.0 nodes** and **49.0 to 22.5 ms**, with refusal correctness unchanged. Chunk size is a flat curve (hit@4 saturated); smaller chunks sharpen top-1 similarity at the cost of a larger index.

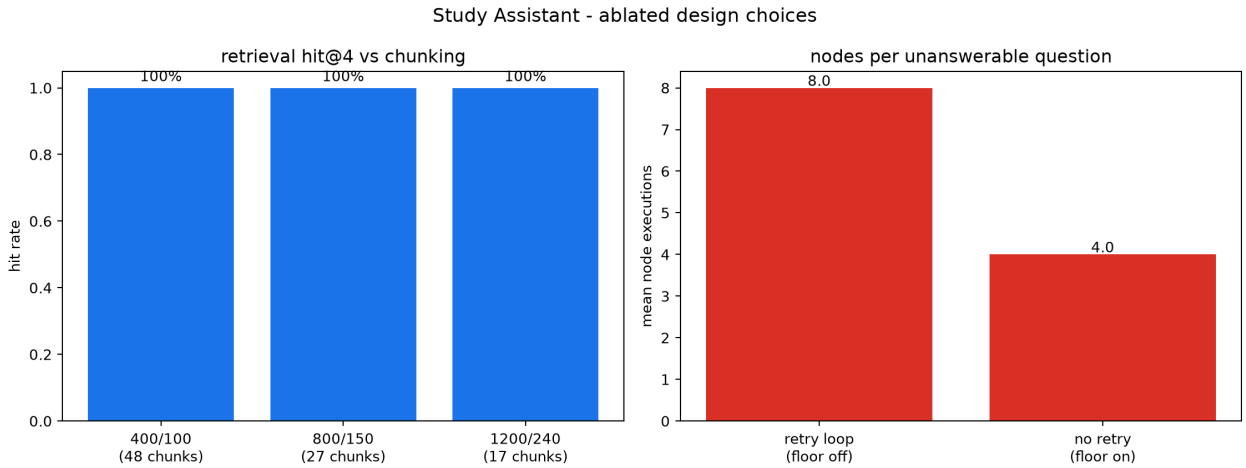


Figure 2. Chunking and retry-policy ablations.

5. Traces

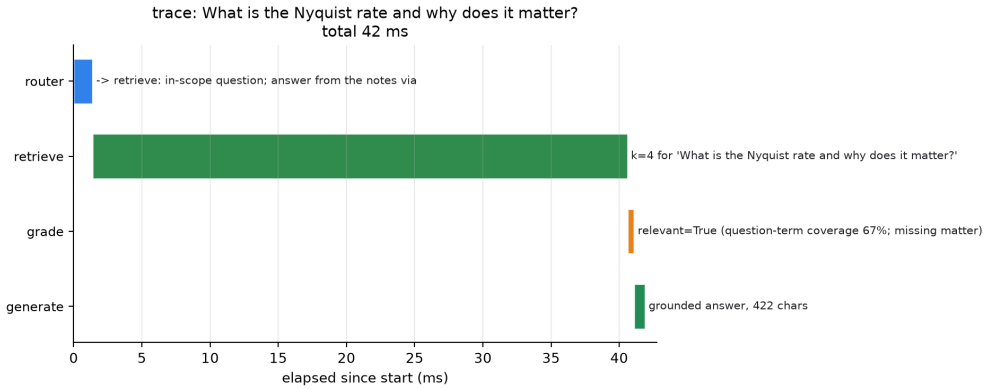


Figure. grounded answer, happy path (local render).

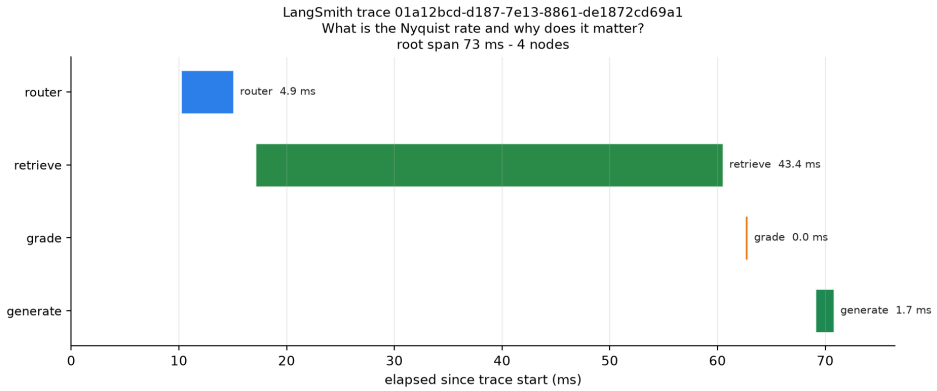


Figure. the same path, from the live LangSmith trace.

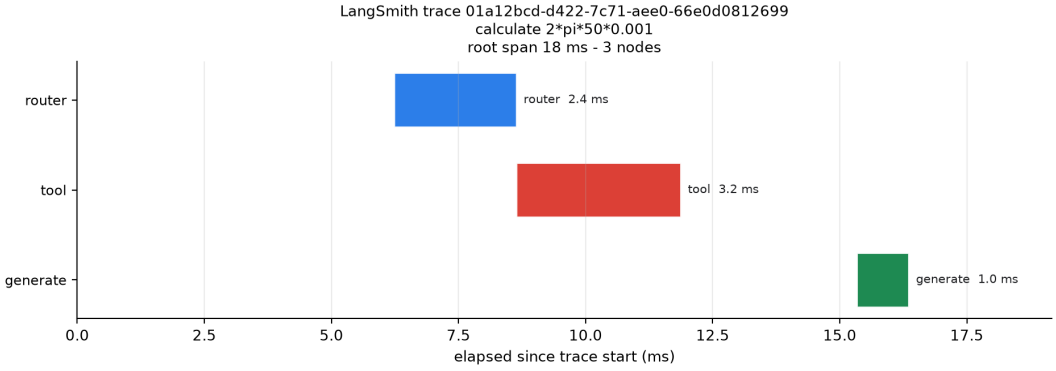


Figure. tool path from the live LangSmith trace.

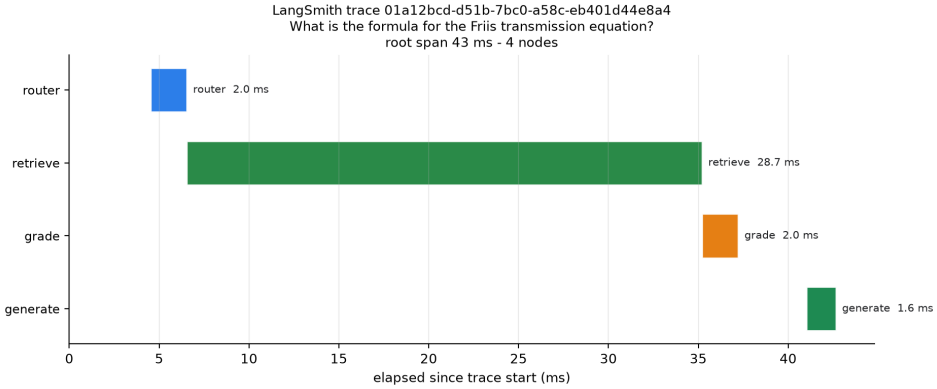


Figure. the q20 Friis failure, from LangSmith.

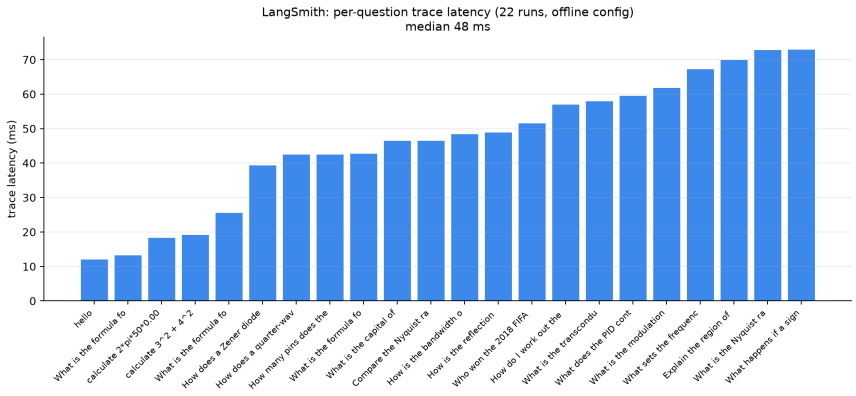


Figure. per-question trace latency (LangSmith).

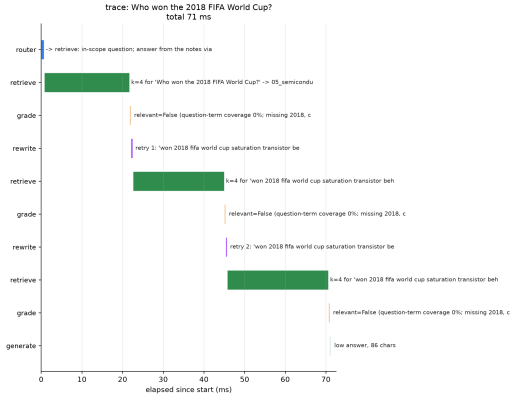


Figure. retry loop (floor off) on an unanswerable question.

6. Demo (4.4)

src/app.py is a Streamlit chat UI showing the answer, a confidence badge, the citations, and an expandable full agent trace and retrieved chunks for every turn, with the live backend and index stats in the sidebar.

Live demo: pending deployment

7. Limitations

- Small self-authored corpus and test set; the 100% metrics are ceilings, not estimates.
- A lexical relevance grader - the direct cause of the headline failure (q20).
- Extractive default answering; the Groq generative baseline (4.3) matches the offline routing/retrieval and fixes the two weak metrics (refusal 67->100%, keyword 93->100%) at a median ~4.4 s vs 56 ms.
- Single-run measurements; the hosted router/grader vary slightly run to run (q10), and the measured pass used gpt-oss-20b rather than the configured gpt-oss-120b.

8. Reproduce

python data/build_corpus.py; python src/ingest.py; python eval/run_eval.py; python eval/improvement_experiment.py; streamlit run src/app.py. Set GROQ_API_KEY, GOOGLE_API_KEY or LANGSMITH_API_KEY in .env to enable the hosted model and remote tracing.